

GENESIS AI ENGINEERING HANDBOOK

SPRINT 1B – DELIVERABLE EF-03

ENTERPRISE ARCHITECTURE SPECIFICATION

Project: Genesis AI

Product: Genesis Product Factory

Version: 1.0

Status: Draft

Chapter 1 — Purpose

This document defines the complete enterprise architecture of Genesis AI.

It specifies how every system component, department, AI employee, workflow, database, and external integration interacts.

This document is the highest technical authority after the Genesis Constitution.

Chapter 2 — Architectural Principles

Genesis follows Enterprise Architecture principles.

EA-001

Single Source of Truth

Business data exists only in the Genesis Database.

EA-002

Loose Coupling

Departments communicate through reports.

Never direct dependencies.

EA-003

Single Responsibility

Every AI Employee owns exactly one responsibility.

EA-004

Human Governance

Humans approve strategy.

AI executes operations.

EA-005

Provider Independence

Genesis must never depend on one marketplace, AI model, or infrastructure provider.

EA-006

Version Everything

Every report, workflow, prompt, document, and decision is versioned.

Chapter 3 — Enterprise Layers

Genesis contains nine logical layers.

Layer 1

Founder

↓

Layer 2

Founder Dashboard

↓

Layer 3

Genesis Orchestrator



Layer 4

Departments



Layer 5

AI Employees



Layer 6

Knowledge Engine



Layer 7

Data Layer



Layer 8

Integration Layer



Layer 9

External Platforms

Each layer may only communicate with adjacent layers.

Chapter 4 — System Components

Genesis consists of the following components.

Genesis Orchestrator

Workflow Engine

Queue Manager

Approval Engine

Research Department

Product Department

Creative Department

Marketing Department

Publishing Department

Analytics Department

Knowledge Engine

Database

External Connectors

Notification Engine

Audit Engine

Version Engine

Future Plugin Manager

Chapter 5 — Genesis Orchestrator

Mission

Coordinate workflows.

Responsibilities

Workflow creation

Department assignment

Approval routing

Failure handling

Status tracking

Retry scheduling

The Orchestrator does NOT perform business work.

It coordinates departments.

Chapter 6 — Workflow Engine

Every request becomes a workflow.

Workflow Lifecycle

Created

↓

Queued

↓

Running

↓

Waiting Approval

↓

Completed

↓

Archived

Possible Alternate States

Rejected

Paused

Retry

Cancelled

Failed

Chapter 7 — Queue Architecture

Every department owns a queue.

Research Queue

↓

Product Queue

↓

Creative Queue

↓

Marketing Queue

↓

Publishing Queue

↓

Analytics Queue

Queues isolate failures.

Queues allow future parallel execution.

Chapter 8 — Event Architecture

Genesis is Event Driven.

Standard Events

Project Created

Research Started

Research Completed

Research Failed

CEO Review Requested

CEO Approved

CEO Rejected

Blueprint Created

Creative Started

Creative Completed

Marketing Started

Marketing Completed

Publishing Ready

Published

Analytics Updated

Events are immutable.

Events generate audit records.

Chapter 9 — Report Architecture

Departments exchange reports instead of conversations.

Research Report

↓

Product Blueprint

↓

Creative Pack

↓

Marketing Pack

↓

Publishing Package

↓

Business Performance Report

Reports become permanent business assets.

Chapter 10 — AI Memory Architecture

Genesis contains five memory layers.

Working Memory

Current task only.

Destroyed after completion.

Department Memory

Stores temporary department context.

Example

Research history

Creative iterations

Marketing drafts

Business Memory

Stores everything related to one business.

Brand

Products

Research

Marketing

Analytics

Organizational Memory

Stores reusable assets.

Templates

Prompts

Guidelines

Brand Systems

Lessons Learned

Learning Memory

Stores optimization knowledge.

Winning prompts

Best performing niches

High conversion products

Successful strategies

Future AI employees learn from this layer.

Chapter 11 — State Machines

Every entity follows defined states.

Project

Created

↓

Research

↓

Approved

↓

Planning



Creative



Marketing



Publishing



Analytics



Completed



Archived

No undefined transitions permitted.

Chapter 12 — Sequence Architecture

Example

Founder submits project.



Orchestrator creates workflow.



Research Queue.



Research Department.

↓

Research Report.

↓

CEO Decision.

↓

Product Queue.

↓

Creative Queue.

↓

Marketing Queue.

↓

Publishing Queue.

↓

Founder Review.

↓

Release.

Chapter 13 — Failure Architecture

Possible failures.

Research Failure

Creative Failure

Publishing Failure

Database Failure

Integration Failure

Workflow Failure

Each failure produces.

Error Code

Timestamp

Department

Workflow ID

Recovery Strategy

Retry Count

Owner

Chapter 14 — Retry Strategy

Automatic retries.

Research

3

Creative

2

Publishing

5

Notifications

Unlimited

Permanent failures require Founder review.

Chapter 15 — Knowledge Architecture

Knowledge is separated into domains.

Research

Creative

Marketing

Products

Prompts

Architecture

Business

Engineering

Knowledge is searchable.

Knowledge is reusable.

Knowledge is versioned.

Chapter 16 — Security Architecture

Authentication

Authorization

Audit

Encryption

Secret Management

Role Separation

Least Privilege

No employee receives unnecessary permissions.

Chapter 17 — Observability

Genesis records.

Execution Time

Queue Length

Department KPIs

Employee KPIs

Errors

Retries

Business KPIs

API Calls

Cost

Latency

Everything is measurable.

Chapter 18 — Scalability Strategy

Genesis scales in four directions.

More Businesses

More Departments

More Employees

More Integrations

Scaling should require configuration rather than redesign.

Chapter 19 — Plugin Architecture

Future providers become plugins.

Examples

Qikink Plugin

Amazon Plugin

Shopify Plugin

Meta Plugin

WhatsApp Plugin

Pinterest Plugin

Removing one plugin must not affect other modules.

Chapter 20 — Multi-Tenant Architecture

Future Genesis supports multiple businesses.

Business A

↓

Departments

↓

Database

↓

Reports

Business B remains isolated.

Shared knowledge is configurable.

Chapter 21 — Deployment Architecture

Phase 1

ChatGPT

Google Sheets

Google Drive

↓

Phase 2

Supabase

n8n

Cloudflare R2



Phase 3

Dashboard

Authentication

Plugins



Phase 4

BusinessOS

Distributed Workers

Chapter 22 — Architecture Decision Records (ADR)

Every architectural decision requires an ADR.

ADR Structure

Decision

Problem

Alternatives

Reasoning

Consequences

Future Review Date

No undocumented architecture changes.

Chapter 23 — Engineering Standards

Every component must satisfy.

Single Responsibility

Loose Coupling

High Cohesion

Idempotency

Observability

Version Control

Documentation

Auditability

Provider Independence

Chapter 24 — Future Enterprise Modules

Finance

CRM

HR

Inventory

Manufacturing

Legal

Vendor Management

Business Intelligence

Supply Chain

The architecture reserves interfaces for these modules.

CTO Review

Overall Assessment

This architecture is now suitable as the enterprise blueprint for Genesis.

Remaining Improvements (Enterprise Edition)

Before implementation begins, I recommend creating:

1. **Component diagrams**
2. **Sequence diagrams**
3. **Entity-Relationship (ER) diagrams**
4. **Deployment diagrams**
5. **C4 architecture model (Context, Container, Component, Code)**
6. **OpenAPI specifications for every service**
7. **Formal state diagrams for each workflow**

These additions will make the architecture documentation comparable to that used in large-scale enterprise software projects.

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Sprint 1B Progress: 3 / 8 Engineering Deliverables Complete